Simulations of Turbulent Boundary Layers with Heat Transfer

by

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Abstract

A new parallelisation of the existing fully spectral research code has been implemented and validated, and used to perform simulations on massively parallel computer architectures with $\mathcal{O}(1000)$ cores. Using the parallelised code, direct numerical simulations (DNS) and large-eddy simulations (LES) of a spatially developing turbulent boundary layer with and without passive scalars over a flat plate under zero-pressure gradient (ZPG) have been carried out. The Navier-Stokes equations are solved employing a spectral method with up to 600 M grid points. The Reynolds numbers obtained are the highest for a turbulent boundary layer obtained to date with such a numerical setup. An extensive number of turbulence statistics for both flow and scalar fields are computed and compared to the well-established experimental/numerical database. In general, good agreements are found. Premultiplied spanwise and temporal spectra are also used to identify the large-scale motions in the outer part of the boundary layer. The similarities shared by the streamwise velocity and the scalar with $Pr = 0.71$ indicate that they might be generated by the same mechanism. The effects from the different Prandtl numbers and wall boundary conditions are also discussed in detail. Furthermore, the effects of the free-stream turbulence (FST) on the heat transfer on the wall are examined. This problem is of great interest in industrial applications. The momentum and heat transfer on the wall are compared with those obtained with a clean free stream and augmentations of both momentum and heat transfer in the turbulent region are found. In addition, the boundary layer structures are studied and a change of the structures in the outer region are found due to the presence of the free-stream turbulence. By examining the one-dimensional spanwise spectrum, it is speculated that the increase of the momentum and heat transfer are associated with the large-scale motions in the outer layer.

Descriptors: direct numerical simulation (DNS), large-eddy simulation (LES), turbulent boundary layer, passive scalar, large-scale structures, massively parallel simulations

Preface

This thesis deals with spatially evolving turbulent boundary-layer flows including heat transfer. A brief introduction on the basic concepts and methods is presented in the first part. The second part is a collection of the following articles:

Paper 1. Q. Li, P. Schlatter & D. S. Henningson,
Spectral simulations of wall-bounded flows on massively-parallel computers. Internal report (2008)

Paper 2. P. Schlatter, R. Örlü, Q. Li, G. Brethouwer, J. H. M. Fransson, A. V. Johansson, P. H. Alfredsson & D. S. Henningson,
Turbulent boundary layers up to $Re_\theta = 2500$ *studied through numerical simulation and experiments.* Published in Phys. Fluids **21**, 051702 (2009)

Paper 3. P. Schlatter, Q. Li, G. Brethouwer, A. V. Johansson & D. S. Henningson,
Simulations of high-Reynolds number turbulent boundary layers. Submitted to Int. J. Heat Fluid Flow, invited paper from TSFP-6, Seoul, South Korea, 2009

Paper 4. Q. Li, P. Schlatter, L. Brandt, & D. S. Henningson,
Direct numerical simulation of a spatially developing turbulent boundary layer with passive scalar transport. Int. J. Heat Fluid Flow, in press (2009)

Paper 5. Q. Li, P. Schlatter & D. S. Henningson,
Simulations of heat transfer in a boundary layer subject to free-stream turbulence. Published in TSFP-6, Seoul, South Korea, 2009

Division of work between authors
The research project was initiated by Dr. Philipp Schlatter (PS) who has acted as co-advisor and Prof. Dan S. Henningson (DH) who has been the main supervisor.

Paper 1
The code development was performed by Qiang Li (QL) with the help from PS. The paper has been written by QL with input from PS and DH.

Paper 2
The simulation code validation, problem setup and the performance and optimisation tuning was performed by QL and PS. The simulations were performed by PS and the paper was written by PS with input from the co-authors.

Paper 3
The simulation code validation, problem setup and the performance and optimisation was performed by QL and PS. The computations were performed by PS and the paper was written by PS with input from QL, Dr. Geert Brethouwer, Prof. Arne V. Johansson and DH.

Paper 4
The simulations were performed by QL and the paper has been written by QL with input from PS, Dr. Luca Brandt and DH.

Paper 5
The code adaptation and the computations were performed by QL and the paper was written by QL with input from PS and DH.

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Part I

Introduction

CHAPTER 1

Introduction

Most flows in nature and engineering applications are turbulent and thus characterised by rapid fluctuations in both time and space. Over hundred years have passed and people still do not fully understand turbulence even though it is a universal phenomenon. Examples of such flows include billowing of clouds in the sky, oceanic currents, the flow over the wing of an aircraft or over gas turbine blades. Even the blood flow in our bodies is sometimes turbulent, e.g. the blood in the heart, especially in the left ventricle. Moreover, some flows in everyday life are deliberately made to be turbulent, e.g. coffee stirring, tea pouring and cocktail shaking. Such inherent random and complex characteristics make turbulence to be an intellectual challenge for a great number of scientists and engineers and will continue to pose this challenge in the future. Consequently, a huge amount of theoretical and experimental work has been performed during the last centuries to shed light on this mysterious field.

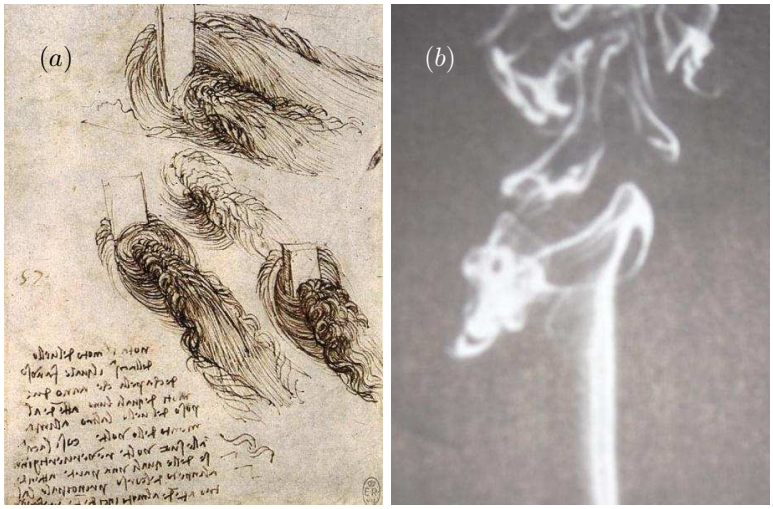


FIGURE 1.1. (*a*) Sketch from Leonardo da Vinci's notebooks. The picture is taken from Geurts (2003). (*b*) Meandering of smoke from a cigarette. The picture is taken from Libby (1996).

Dating back to early 16th century, the Italian genius Leonardo da Vinci sketched the famous "Old Man with Water Studies" (see Figure 1.1 (a)) and the following observations (Piomelli 2001) are made: "The water has eddying motions, one part of which is due to the principal current, the other to the random and reverse motion." These pure observations may be seen as a precursor to the Reynolds' decomposition which is a sophisticated way to study turbulence. The first example of a visualisation of a flow developing from laminar into turbulent motion is the famous experiment carried out by Osborne Reynolds in 1883. He studied the flow of water in a glass tube using ink as a dye. For low flow rates a steady dye stream was observed to follow a straight path through the tube. As the flow rate was increased, at some point, "the colour band would all at once mix up with the surrounding water, and fill the rest of the tube with a mass of coloured water", see e.g. Jackson & Launder (2007). In order to quantify these experimental results, he introduced a non-dimensional number $Re = \frac{UL}{\nu}$, now known as the Reynolds number, in his classic paper (Here U denotes the velocity scale, L the length scale and ν is the kinematic viscosity). Now it is known that the flow will become turbulent in practice, if the Reynolds number exceeds a value about 2000.

Another typical example demonstrating the nature of a turbulent flow is the meandering of the smoke from a cigarette (see Figure 1.1 (b)). The smoke first moves straight upward and suddenly changes to a turbulent status. This experiment not only provides the visualisation of the turbulent motion but also the spreading of a passive scalar in a turbulent flow. A passive scalar is a diffusive contaminant in a fluid flow that, due to its low concentration, has no influence on the fluid motion but is however influenced by the fluid motion. Typical passive scalars are e.g. small amounts of heat, salt in the sea or pollutants in atmospheric or oceanic flows. An understanding and prediction of the passive scalar behaviour in a turbulent flow is crucial since the turbulent momentum and scalar transport play a key role in many engineering applications and will be of growing importance in global environmental problems (Kasagi & Iida 1999).

When fluids flow above a solid body, the fluid elements in a thin layer near the body surface will be retarded due to the effect of friction while the other elements outside this thin layer are not affected. This thin layer is usually called boundary layer. The concept of the boundary layer was first proposed by Ludwig Prandtl in 1905. He hypothesised that the viscous effects are negligible everywhere except in the boundary layer. Furthermore, many interesting parameters of the aerodynamic property of a body, e.g. drag coefficient, are determined in precisely this boundary layer. Thus it is of great engineering significance in most applications to understand and (numerically or experimentally) fully resolve this thin layer of fluid.

Although the phenomena of turbulent flows vary from one to the other, the governing equations describing the motion of the Newtonian fluids are always the same. These equations are the so-called Navier-Stokes (NS) equations,

named after Claude-Louis Navier and George Gabriel Stokes who first formulated them in the 19th century. Together with the continuity equation (conservation of mass) and the scalar transport equation which governs the evolution of the scalar field, a system of equations consisting of five variables (three velocity components, pressure and scalar concentration) is established. However, the full equation system is non-linear and time dependent partial differential equations, and thus no analytical solutions exists except for the non-turbulent cases. Therefore, one has to calculate approximate solutions numerically with the aid of computers as an alternative to experiments.

Thanks to the rapid progress in high-performance computers, direct numerical simulation (DNS) and large-eddy simulation (LES) became important tools for turbulence research in the last couple of decades (Rogallo & Moin 1984; Moin & Mahesh 1998) whereas it is still dominated by the Reynolds-averaged Navier-Stokes (RANS) type of simulations in commercial codes. Additional reviews of the turbulent heat transfer and passive scalars are made by e.g. Kasagi & Iida (1999) and Warhaft (2000). DNS provides three-dimensional, time-dependent numerical solutions to the NS equations and the scalar transport equation. These equations are supposed to be solved as accurately as possible without employing any turbulence models, hence the errors come only from the numerical approximations and domain truncation. On the other hand, LES solves the equations with less grid points but supplemented with a so-called sub-grid scale (SGS) model to compensate for the omitted grid points. The overall performance of a LES calculation thus to a large extent depends on the chosen SGS model assuming sufficient grid points to resolve the relevant physics. RANS only gives averaged solutions to the Navier-Stokes equations with the so-called $k - \varepsilon$ model being most commonly used. One may refer to Pope (2000) for more detailed discussions about RANS models.

One important feature of turbulent flow is that it consists of temporal and spatial scales with a broad spectrum. The smallest scale arising in turbulent flows is usually the so-called Kolmogorov microscale η which is determined by the flow properties itself while the largest scales are commonly restricted by the flow geometry, e.g. the channel height or the boundary-layer thickness. A simple dimensional scaling argument leads to the conclusion that the resolution (number of grid points) of a DNS calculation required in each direction is proportional to $Re^{\frac{3}{4}}$ (Piomelli 2001; Jiménez 2003) and the exponent can increase up to 0.9 depending on the definitions of the Reynolds number (Reynolds 1989). The total computational time is then roughly proportional to Re^3 (Geurts 2003). This means a doubling of the Reynolds number would be feasible every 6 to 7 years if the rate of increase of the computer performance was maintained (Sandham 2002). Up to now, it seems that there is no signs of the computational capacity to slow down and especially recently there is a trend of using massively parallel processing (MPP) system, e.g. the Blue Gene systems, for large calculations.

For wall-bounded flows, the resolution requirement of a well-resolved LES calculation is that the number of grid points has to be sufficient to resolve the wall layer according to Piomelli (2001), i.e. to capture the energy-producing structures, e.g. the so-called streaks. The cost of the LES calculations also depends on the Reynolds number. As estimated by Chapman (1979), the resolution required to resolve the viscous sub-layer ($y^+ < 5$) is proportional to $Re^{1.8}$. The + denotes the wall scaling, i.e. scaled by $\frac{\nu}{u_\tau}$ where $u_\tau = \sqrt{\frac{\tau_w}{\rho}}$ is the friction velocity and τ_w the wall shear stress. Jiménez (2003) made a similar estimation that the resolution is proportional to Re_τ^2 where Re_τ is the Reynolds number based on the friction velocity u_τ. This means that using LES one can obtain results at higher Reynolds number than DNS for equivalent cost, but the Reynolds number reached is still not high enough for practical engineering flows. Recent studies have shown especially for wall-bounded turbulent flows, that a good LES is almost as expansive as a DNS, e.g. see the discussion in Fröhlich & Rodi (2002). Therefore, one has to resort to the wall models (Piomelli & Balaras 2002) or hybrid LES/RANS or detached eddy simulation (DES) (Spalart 2009) approach to perform calculations in an acceptable time for practical engineering applications. This in turn requires a good RANS model.

For the DNS of scalar transfer in turbulent flows, not only the Reynolds number but also the molecular Prandtl number (Pr) has to be considered. The molecular Prandtl number is a dimensionless number which is a measure of the ratio between viscous diffusion and scalar diffusion. Typical values for Pr are 0.71 for air, around 7 for water, between 100 and 40,000 for engine oil and around 0.025 for mercury (White 2006). For turbulent flows with scalar transport, the smallest scales in the scalar fluctuations decrease with the increase of Pr which makes the calculation even more expensive. The ratio of the largest to the smallest scales is approximately proportional to $Re^{\frac{3}{4}}Pr^{\frac{1}{2}}$ at very high Pr (Batchelor 1959; Tennekes & Lumley 1972).

In 1960s, the first LES calculations were performed by meteorologists (Smagorinsky 1963; Deardorff 1970) to predict atmospheric flows. A brief history of LES can be found in Lesier *et al.* (2005). The first DNS of turbulent flow starts with decaying isotropic turbulence using spectral methods, see the historical account by Canuto *et al.* (1988), on a 32^3 grid by Orszag & Patterson (1972). Later, Kim et al. (1987) simulated a fully developed channel flow using Fourier/Chebyshev algorithms at $Re_\tau = 180$ where Re_τ is the Reynolds number based on the friction velocity u_τ and the channel half width h. The first simulation of a fully turbulent boundary layer was performed by Spalart (1988) using an innovative spatio-temporal approach at $Re_\theta = 300, 670, 1410$ where the Reynolds number Re_θ is based on the free-stream velocity U_∞ and momentum thickness θ. The first DNS with passive scalars was performed by Rogers et al. (1986) in a homogeneous shear flow and by Kim and Moin (1989) in a channel flow of $Pr = 0.1$, 0.71 and 2.0 at $Re_\tau = 180$. DNS of turbulent

boundary layer were performed for the first time by Bell and Ferziger (1993) with $Pr = 0.1$, 0.71 and 2.0 up to a medium Reynolds number of $Re_\theta = 700$.

For canonical geometries, such as channel or flat-plate boundary layer, usually highly accurate and efficient algorithms, e.g. spectral methods (Canuto *et al.* 1988) which have barely no dissipation and dispersion errors, can be employed. However, spectral methods have difficulties in handling complex geometries. One way to overcome this drawback is to use the so-called spectral-element method first introduced by Patera (1984) and later developed by many others, see e.g. Maday & Patera (1989); Karniadakis & Sherwin (2005). The latter method combines the geometric generality/flexibility of the finite element method with the accuracy of spectral methods. However, the running cost, inefficient algorithms and stability issues have limited the applications of this method so far mainly to laminar and early transitional flows. Other possibilities are the use of the low-order schemes with extra grid points to compensate for the low accuracy. See more discussions in Sandham (2002).

One unique feature of numerical simulations is that they are able to simulate "wrong" physics as mentioned in Jiménez & Pinelli (1999), e.g. such as spurious modes, wrong thresholds, incorrect reattachment points, underestimated/overestimated functions due to under-resolved simulations, irrelevant size of the computational box, not appropriate boundary conditions or unrealistic parameters, etc. Thus, to generate simulation data with good quality is not simple or cheap at all. Therefore, the validation of the results becomes relatively important. Suggestions made in Sandham (2002) for DNS calculations are summarised here:

- validate the code against analytical solutions (if possible), asymptotic limits and well-established database;
- carry out systematic studies by varying the resolutions, box sizes, time steps and numerical methods.

For LES, Geurts (1999) interpreted it as a balancing between different sources of errors, i.e. modelling and numerical ones. This makes the validation of LES results even more complicated. In addition to the guidelines suggested for DNS, one should also follow the rules (Geurts & Leonard 2002)

- avoid dissipative numerical methods;
- vary the filter width while fixing the resolution;
- use dynamic modelling.

Moreover, to validate a given SGS model, one could conduct either *a priori* or *a posteriori* tests during which accurate data from either DNS or experiments are required. In *a priori* test, no actual simulations are carried out. The SGS model is evaluated based on the available DNS or experimental data and compared with the exact SGS stress. However, care has to be taken when interpreting the results since a successful *a priori* test does not guarantee good results in an actual LES calculation; sometimes the simulations can be even unstable (Vreman *et al.* 1995). On the other hand, a poor *a priori* test does not

necessarily lead to poor results in an actual run (Meneveau 1994). Therefore, to judge whether a SGS model performs well or not, one has to run a simulation with the correctly implemented model and then compare the results with DNS or experimental data. This approach is known as *a posteriori* test (Piomelli *et al.* 1988) in the LES community. To correctly compare the obtained LES results with DNS or experimental database, one should refer to e.g. Winckelmans *et al.* (2002); Sagaut (2005) for more detail.

CHAPTER 2

Theoretical formulation

2.1. Governing equations

2.1.1. *Direct numerical simulation (DNS)*

The governing equations for the direct numerical simulation (DNS) of an incompressible (laminar, transitional or turbulent) flow with passive scalars are the Navier-Stokes equations and the scalar transport equation, here written in non-dimensional form and tensor notation as

$$\begin{aligned} \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} &= -\frac{\partial p}{\partial x_i} + \frac{1}{Re} \frac{\partial^2 u_i}{\partial x_j \partial x_j} \ , \\ \frac{\partial u_i}{\partial x_i} &= 0 \ , \\ \frac{\partial \theta}{\partial t} + u_i \frac{\partial \theta}{\partial x_i} &= \frac{1}{RePr} \frac{\partial^2 \theta}{\partial x_i \partial x_i} \ , \end{aligned} \tag{2.1}$$

where $(x_1, x_2, x_3) = (x, y, z)$ are the Cartesian coordinates in the streamwise, wall-normal and spanwise direction, respectively. $(u_1, u_2, u_3) = (u, v, w)$ are the corresponding instantaneous velocity fields, p is the pressure and θ the scalar concentration. The Reynolds number Re is defined as

$$Re = \frac{U_{\mathrm{ref}} L_{\mathrm{ref}}}{\nu} \ ,$$

where U_{ref}, L_{ref} and ν are the dimensional reference velocity and length and kinematic viscosity, respectively, which are also used for non-dimensionalisation. Pr is the Prandtl number defined by

$$Pr = \frac{\nu}{\alpha} \ ,$$

where α is the scalar diffusivity.

The summation convention is implied over repeated indices throughout this thesis unless stated otherwise. The streamwise and spanwise directions will be alternatively termed as the horizontal/wall-parallel directions.

2.1.2. *Large-eddy simulation (LES)*

The governing equations for large-eddy simulation (LES) are the spatially filtered Navier-Stokes equations and the scalar transport equation which in dimensionless form read

$$\begin{aligned} \frac{\partial \widetilde{u}_i}{\partial t} + \widetilde{u}_j \frac{\partial \widetilde{u}_i}{\partial x_j} &= -\frac{\partial \widetilde{p}}{\partial x_i} + \frac{1}{Re} \frac{\partial^2 \widetilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} \ , \\ \frac{\partial \widetilde{u}_i}{\partial x_i} &= 0 \ , \\ \frac{\partial \widetilde{\theta}}{\partial t} + \widetilde{u}_i \frac{\partial \widetilde{\theta}}{\partial x_i} &= \frac{1}{RePr} \frac{\partial^2 \widetilde{\theta}}{\partial x_i \partial x_i} - \frac{\partial \sigma_i}{\partial x_i} \ , \end{aligned} \tag{2.2}$$

where the so-called SGS stresses $\tau_{ij} = \widetilde{u_i u_j} - \widetilde{u}_i \widetilde{u}_j$ and $\sigma_i = \widetilde{u_i \theta} - \widetilde{u}_i \widetilde{\theta}$ represent the impact of the unresolved velocities/scalar on the resolved ones and have to be modelled. Mathematically, they arise from the nonlinearity of the convection term which does not commute with the linear filtering operation.

A filtered quantity $\widetilde{f}(x)$ in one dimension is defined by

$$\widetilde{f}(x) = G^P * f = \int_\Omega G^P(x, x', \Delta) f(x') \mathrm{d}x' \ ,$$

where Ω is the computational domain, G^P is the primary filter with Δ being the filter width. A three-dimensional filter can be easily formulated and the common filters are the sharp cut-off filter, Gaussian filter and top-hat filter. For more mathematical properties of these filters, one could refer to e.g. Pope (2000). The filtering operation separates the large and small scales and the filter function G^P determines the size and structures of the large scales. For the differentiation and the filtering operation to commute, G^P has to be a function of $x - x'$ only (Leonard 1974). A rough classification of the LES models can be made depending on the primary filter G^P: implicitly filtered models if the primary filter G^P is the grid filter, e.g. the ones used in the classical eddy-viscosity models; explicitly filtered models where an explicit filter operation is taken, e.g. the graded primary filter used in the approximate deconvolution model (ADM) (Stolz & Adams 1999).

2.2. Sub-grid scale modelling

In LES, the effects of the unresolved scales on the resolved ones have to be provided via the sub-grid scale (SGS) model. Therefore, a good SGS model is in generally desirable.

2.2.1. *Smagorinsky model (SM)*

The Smagorinsky model proposed first by Smagorinsky (1963) is the most widely used SGS model today. It belongs to the class of eddy-viscosity models since the traceless/deviatoric part of the SGS stresses τ_{ij} is modelled as

$$\tau_{ij} - \frac{\delta_{ij}}{3} \tau_{kk} = -2\nu_t \widetilde{S}_{ij} \ ,$$

where δ_{ij} is the Kronecker delta, defined by

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} .$$

The strain rate of the resolved velocity field $\widetilde{S}_{ij}$ is calculated according to

$$\widetilde{S}_{ij} = \frac{1}{2}\left(\frac{\partial \widetilde{u}_i}{\partial x_j} + \frac{\partial \widetilde{u}_j}{\partial x_i}\right) .$$

The eddy viscosity ν_t is obtained by

$$\nu_t = (C_s\Delta)^2|\widetilde{S}| ,$$

where $|\widetilde{S}|$ is defined by

$$|\widetilde{S}| = \sqrt{2\widetilde{S}_{ij}\widetilde{S}_{ij}} .$$

Δ is a typical length scale of the primary filter usually computed from the grid size as $\Delta = (\Delta x \Delta y \Delta z)^{\frac{1}{3}}$ (Deardorff 1970). The model coefficient C_s known as the Smagorinsky constant is actually not a constant and has to be determined *a priori* based on the flow case. Lilly (1967) derived an approximate value for the Smagorinsky constant $C_s \approx 0.18$ in homogeneous isotropic turbulence. However, this value is found to cause excessive damping in wall-bounded flows and has to be reduced to $C_s \approx 0.1$ to sustain turbulence as reported by Deardorff (1970); Moin & Kim (1982); Piomelli *et al.* (1988). McMillan *et al.* (1980) confirmed in *a priori* tests that in homogeneous turbulence C_s does decrease with increasing strain rate.

Despite that SM with constant coefficient is successful in certain flow types, it suffers from a number of drawbacks. One major drawback is that the sub-grid scale stress does not vanish close to solid walls. One remedy is to use the so-called van Driest wall damping function (van Driest 1956). Other drawbacks are that e.g. the model does not predict the correct growth rate of the initial disturbances due to excessive damping and it does not account for the so-called "backscatter", i.e. energy transfer from unresolved scales to resolved scales, which in certain circumstances can be significant (Piomelli *et al.* 1991; Härtel *et al.* 1994). The capability of accounting for backscatter is usually considered as a merit of a LES model.

2.2.2. *High-pass filtered Smagorinsky model (HPF-SM)*

Alternatively, high-pass filtered (HFP) velocities can be used instead of the low-pass filtered ones to calculate the SGS model contribution. Ducros *et al.* (1996) showed that the filtered structure function (FSF) model based on the structure function (SF) model proposed by Métais & Lesieur (1992) was able to predict transitional flow. Later, Sagaut *et al.* (2000) employed a high-pass filtered velocity to calculate the eddy viscosity. Not only the eddy viscosity, but also the strain rate could be calculated based on high-pass filtered quantities. These models were first proposed by Vreman (2003) and Stolz *et al.* (2005) based on SM closure. With the high-pass filtered velocities $H * \widetilde{u}$ where H is a

high-pass filter, the computation of the SGS stress τ_{ij}, the strain rate $S_{ij}(H*\widetilde{u})$ and the eddy viscosity ν_t^{HPF} become

$$\begin{aligned}
\tau_{ij} - \frac{\delta_{ij}}{3}\tau_{kk} &= -2\nu_t^{HPF} S_{ij}(H*\widetilde{u}) \ , \\
S_{ij}(H*\widetilde{u}) &= \frac{1}{2}\left(\frac{\partial H*\widetilde{u}_i}{\partial x_j} + \frac{\partial H*\widetilde{u}_j}{\partial x_i}\right) \ , \\
\nu_t^{HPF} &= (C_s^{HPF}\Delta)^2 |S(H*\widetilde{u})| \ , \\
|S(H*\widetilde{u})| &= \sqrt{2S_{ij}(H*\widetilde{u})S_{ij}(H*\widetilde{u})} \ .
\end{aligned}$$

There is a close relation between the HPF models and the variational multiscale (VMS) method presented by Hughes *et al.* (2000, 2001). The two methods differ in how the scale separation is treated. In the HPF methodology, the filtering approach is adopted while in the VMS method, the scale separation is performed by projection onto disjunct functional spaces.

One advantage of the HPF-SM is that the model contribution vanishes on solid boundaries so that no near-wall damping is needed. Moreover, the ability to handle transitional flows is shown in Stolz *et al.* (2005) and it accounts for backscatter (Schlatter *et al.* 2005). However, the model constant C_s^{HPF} does not have the correct asymptotic behaviour which gives rise to restricted time steps (Jeanmart & Winckelmans 2007). Therefore, more general remedies other than the ad-hoc near-wall damping have been proposed. This leads to the introduction of the wall-adapting local eddy viscosity (WALE) model (Nicoud & Ducros 1999) and the dynamic procedure (Germano *et al.* 1991; Lilly 1992).

Using the square of the velocity tensor, the WALE model evaluates the viscosity in such a way that it has the proper near-wall behaviour. However, the more widely used method is the dynamic procedure first proposed by Germano *et al.* (1991) and later refined by Lilly (1992). The idea is to use the local flow information to calculate the model coefficient rather than determine it *a priori*. Here SM will be used as an example to show the formulation of the dynamic procedure.

Filtering the LES equations (2.2) by a second filter, the so-called test filter, denoted by $\widehat{\ }$, one obtains

$$\frac{\partial \widehat{\widetilde{u}}_i}{\partial t} + \frac{\partial \widehat{\widetilde{u}}_i \widehat{\widetilde{u}}_j}{\partial x_j} = -\frac{\partial \widehat{\widetilde{p}}}{\partial x_i} + \frac{1}{Re}\frac{\partial^2 \widehat{\widetilde{u}}_i}{\partial x_j \partial x_j} - \frac{\partial T_{ij}}{\partial x_j} \ , \tag{2.3}$$

where $T_{ij} = \widehat{\widetilde{u}_i \widetilde{u}_j} - \widehat{\widetilde{u}}_i \widehat{\widetilde{u}}_j$ is the SGS stress at the test filter level. The test filter is usually chosen to be a sharp cut-off filter applied only in wall-parallel planes leaving the wall-normal direction unfiltered. Although not fully justified, this two-dimensional filtering approach is widely used in LES community. The test filter width is usually set to $\widehat{\Delta} = 2\Delta$. Different widths have been tested in channel flow simulations but no noticeable difference has been reported (Germano *et al.* 1991). By simple mathematical manipulations of equations (2.2) and (2.3), the so-called Germano identity (Germano *et al.* 1991; Germano 1992)

is recovered

$$L_{ij} = \widehat{\tau}_{ij} - T_{ij} \tag{2.4}$$

where $L_{ij} = \widehat{\widetilde{u}_i}\widehat{\widetilde{u}}_j - \widehat{\widetilde{u}_i\widetilde{u}_j}$ is the so-called resolved turbulent stress (Germano *et al.* 1991; Lilly 1992). Applying the Smagorinsky closure to both τ_{ij} and T_{ij}, one obtains

$$\tau_{ij} - \frac{\delta_{ij}}{3}\tau_{kk} = -2(C_s\Delta)^2|\widetilde{S}|\widetilde{S}_{ij} \ ,$$

$$T_{ij} - \frac{\delta_{ij}}{3}T_{kk} = -2(C_s\widehat{\Delta})^2|\widehat{\widetilde{S}}|\widehat{\widetilde{S}}_{ij} \ .$$

Inserting into the Germano identity (2.4) yields

$$L_{ij} - \frac{\delta_{ij}}{3}L_{kk} = 2(C_s\widehat{\Delta})^2|\widehat{\widetilde{S}}|\widehat{\widetilde{S}}_{ij} - 2(C_s\Delta)^2\widehat{|\widetilde{S}|\widetilde{S}_{ij}} = 2C_{dyn}M_{ij} \tag{2.5}$$

where $M_{ij} = \widehat{\Delta}^2|\widehat{\widetilde{S}}|\widehat{\widetilde{S}}_{ij} - \Delta^2\widehat{|\widetilde{S}|\widetilde{S}_{ij}}$ is used as an abbreviation and C_s^2 is replaced by C_{dyn}. In Germano *et al.* (1991), equations (2.5) were contracted with the resolved strain rate $\widetilde{S}_{ij}$ to determine C_{dyn} without physical justification. Note that C_{dyn} solved in this way can become negative and this will account for the backscatter. Lilly (1992) suggested a least-squares approach to minimise the residual $Q = (L_{ij} - \frac{\delta_{ij}}{3}L_{kk} - 2C_{dyn}M_{ij})^2$ and this leads to

$$C_{dyn} = \frac{1}{2}\frac{M_{ij}L_{ij}}{M_{lm}M_{lm}} \ .$$

In practise, averaging in homogeneous direction is usually applied to smoothen C_{dyn} which is considerably fluctuating otherwise. However, this will limit the dynamic model to flows with at least one homogeneous direction.

The dynamic model formulation for the HPF-SM follows the same recipe and will not be repeated here. One may refer to Bruhn (2006); Schlatter *et al.* (2006*a*) for more details.

2.2.3. *Inverse model*

It was realised some time ago that, to some extent, an accurate approximation of a sub-grid scale quantity can be reconstructed from the resolved quantities. The first model based on this idea is introduced by Leonard (1974). Later, the so-called tensor-diffusivity model (TDM) was proposed where the filtered non-linear term is expressed in filtered quantities using Taylor expansions (Leonard & Winckelmans 1999; Winckelmans *et al.* 2001). The SGS stress τ_{ij} with uniform filter width Δ then reads

$$\tau_{ij} = \Delta^2\frac{\partial\widetilde{u}_i}{\partial x_k}\frac{\partial\widetilde{u}_j}{\partial x_k} + \frac{\Delta^4}{2!}\frac{\partial^2\widetilde{u}_i}{\partial x_k\partial x_l}\frac{\partial^2\widetilde{u}_j}{\partial x_k\partial x_l} + \frac{\Delta^6}{3!}\frac{\partial^3\widetilde{u}_i}{\partial x_k\partial x_l\partial x_m}\frac{\partial^3\widetilde{u}_j}{\partial x_k\partial x_l\partial x_m} + \cdots \ ,$$

where summation is assumed over triply repeated indices. For more general filters, one could refer to e.g. Carati *et al.* (2001). If the expansion is truncated after the second-order term, one recovers the so-called gradient model (Clark

et al. 1979). However, these SGS models tend to provide not enough dissipation, therefore, a Smagorinksy eddy-viscosity term is usually supplemented as a regularisation.

Another approach sharing the same idea is the so-called approximate deconvolution model (ADM) which was proposed by Stolz & Adams (1999) for both incompressible and compressible flows (Stolz *et al.* 2001*a*,*b*; Stolz & Adams 2003). Following the notation used in Stolz *et al.* (2001*a*), the approximation of the unfiltered velocity u_i is denoted by $u_i^\star$ and the SGS stress is modelled by

$$\tau_{ij} = \widetilde{u_i^\star u_j^\star} - \widetilde{u}_i \widetilde{u}_j \ ,$$

where the approximately deconvolved quantities $u_i^\star$ is defined by

$$u_i^\star = Q_N * \widetilde{u}_i \tag{2.6}$$

with

$$Q_N = \sum_{\nu=1}^{N} (I - G)^\nu \approx G^{-1} \ , \tag{2.7}$$

where I is the identity operator, G the primary filter and Q_N the approximate inverse of G. N is usually set to 5 and results obtained by using $N > 5$ do not give significant improvements (Stolz *et al.* 2001*a*). Combining equations (2.6) and (2.7), $u_i^\star$ can be approximated by

$$u_i^\star = Q_N * \widetilde{u}_i = 3\widetilde{u}_i - 3\widetilde{\widetilde{u}}_i + \widetilde{\widetilde{\widetilde{u}}}_i + \cdots \ .$$

For high level deconvolution, the van Cittert iterative method used in the ADM is more efficient than evaluating the higher-order terms in the direct reconstruction series as in the TDM (Winckelmans *et al.* 2001). Furthermore, a relaxation regularisation is also needed for ADM which models the interaction of the resolved scales and those not represented on the numerical grid.

Schlatter *et al.* (2006,*b*) showed that for simulations using spectral methods the deconvolution operation applied in the ADM approach is usually not necessary. This leads to a promising and simple LES model, i.e. the relaxation-term (RT) model. In this model, the 3D primary filter only applies to the relaxation term while the non-linear terms are directly computed from the resolved quantities. The SGS forcing then simply reads

$$\frac{\partial \tau_{ij}}{\partial x_j} = \chi (I - Q_N * G) * \widetilde{u}_i = \chi (I - G)^{N+1} * \widetilde{u}_i = \chi H_N * \widetilde{u}_i$$

where $H_N = (I - G)^{N+1}$ is a high-pass filter and χ is the model constant. The constant χ can be determined dynamically, however, tests showed that the results are rather insensitive to it (Stolz *et al.* 2001*a*; Schlatter *et al.* 2006).

The RT model has been shown to obtain accurate results in transitional and turbulent incompressible wall-bounded flows (Schlatter *et al.* 2004, 2006, 2009*a*) and is related to the spectral vanishing viscosity approach by Karamanos & Karniadakis (2000). Since $(I - Q_N * G)$ is positive semidefinite (Stolz *et al.* 2001*a*), the relaxation term is always dissipative and only the small scales

are affected while the large ones are not directly influenced by the model contributions.

2.2.4. *Eddy-diffusivity model*

The sub-grid scale (SGS) model for the passive scalar corresponding to eddy-viscosity models is the so-called eddy-diffusivity model. The SGS stress σ_i is obtained by

$$\sigma_i = -\frac{\nu_t}{Pr_t}\frac{\partial\widetilde{\theta}}{\partial x_i}$$

or

$$\sigma_i = -\frac{\nu_t^{HPF}}{Pr_t}\frac{\partial H*\widetilde{\theta}}{\partial x_i}$$

for classic or high-pass filtered eddy-viscosity models and

$$\sigma_i = -\frac{\chi}{Pr_t}\frac{\partial H_N*\widetilde{\theta}}{\partial x_i}$$

for RT model. A constant turbulent Prandtl number $Pr_t = 0.6$ is usually assumed. Note that all the drawbacks of the models for the velocity are present in the scalar models as well. It should be noted that the literature on SGS modelling of passive scalars is much less extensive than that for the velocities. In particular high Péclet numbers (high Re or Pr) are seldomly addressed.

2.3. Inner and outer scaling of wall turbulence

In many applications, it is the mean flow rather than the rapid fluctuations that is interesting and important. Naturally the Reynolds decomposition of the flow field variables into a mean and a fluctuating quantity is introduced by

$$u_i = \langle u_i\rangle + u_i' \ ,$$
$$p = \langle p\rangle + p' \ ,$$
$$\theta = \langle\theta\rangle + \theta' \ ,$$

where the angular brackets denote a suitably defined mean and a prime denotes the fluctuating part. In a statistically stationary turbulent flow, the mean value of a flow quantity can be taken as the time average or as an average in the homogeneous directions. Generally the mean of a quantity is the ensemble average over an infinite number of realisations. In the present case the mean of a quantity $f(x,y,z,t)$ is defined by

$$\langle f\rangle(x,y) = \lim_{T\to\infty}\frac{1}{T}\int_0^T f(x,y,z,t)\mathrm{d}t \ ,$$

where the spanwise (z) dependence disappears since it is a homogeneous direction.

The behaviour of the flow in the near wall region is of great importance and it was first treated by Ludwig Prandtl. A usual measure of the boundary-layer thickness is δ_{99} which is defined as the wall-normal position at which

the mean streamwise velocity $\langle u \rangle$ reaches the 99% of the free-stream velocity U_∞. However, this is a poorly conditioned quantity both experimentally and numerically, since it depends on the measurement of a small velocity difference (Pope 2000). Therefore, other measures have been proposed based on integral quantities such as the displacement thickness

$$\delta^* = \int_0^\infty \left(1 - \frac{\langle u \rangle}{U_\infty}\right) \mathrm{d}y \ ,$$

and the momentum thickness

$$\theta = \int_0^\infty \frac{\langle u \rangle}{U_\infty} \left(1 - \frac{\langle u \rangle}{U_\infty}\right) \mathrm{d}y \ .$$

The Rotta-Clauser length is also often used and defined by

$$\Delta = \frac{U_\infty \delta^*}{u_\tau} \ .$$

Another quantity known as the shape factor H_{12} which indicates the "fullness" of the mean velocity profile can be defined based on δ^* and θ as

$$H_{12} = \frac{\delta^*}{\theta} \ .$$

For the laminar Blasius boundary layer profile the shape factor $H_{12} = 2.59$ and decreases with increasing Re with an asymptotic value being unity (Nagib et al. 2007).

In the vicinity of the wall, i.e. $y/\delta_{99} \ll 1$, one can show that the total shear stress τ is nearly constant and the viscous effects are important, see e.g. Pope (2000). This is true for all canonical wall-bounded flows, i.e. channel, pipe and turbulent boundary-layer flows. Therefore, the so-called friction velocity u_τ and viscous length scale l_* are chosen to be the velocity and length scales and defined as

$$u_\tau \equiv \sqrt{\frac{\tau_w}{\rho}} = \sqrt{\nu \frac{\partial \langle u \rangle}{\partial y}\Big|_{y=0}} \quad \text{and} \quad l_* = \frac{\nu}{u_\tau} \ ,$$

where ρ is the density.

Using the viscous scalings derived above, the mean streamwise velocity very close to the wall can be written in the form

$$u^+ = \frac{\langle u \rangle}{u_\tau} = \Phi(\frac{y}{l_*}) = \Phi_1(y^+) \ , \tag{2.8}$$

where the "+" denotes normalisation in viscous (wall) units and equation (2.8) is referred to as the law of the wall. Far away from the wall, say $y \sim \delta_{99}$, the viscous effects are not crucial, in contrast, the dynamics are dominated by the turbulence. In this outer region, the so-called velocity-defect law describes the mean streamwise velocity distribution and reads

$$\frac{\langle u \rangle - U_\infty}{u_\tau} = \Psi(\frac{y}{\Delta^*}) = \Psi_1(\eta) \ ,$$

where Δ^* is the outer length scale.

Following the argument by Millikan (1938), a so-called overlap region, i.e. $l_* \ll y \ll \Delta^*$, exists and within which the derivatives of the law of the wall and of the velocity-defect law should asymptoticly match. This yields

$$y^+ \frac{\mathrm{d}\Phi(y^+)}{\mathrm{d}y^+} = \eta \frac{\mathrm{d}\Psi(\eta)}{\mathrm{d}\eta} = \frac{1}{\kappa} \ .$$

This equation can be satisfied only when both sides are constant, which leads to the introduction of the von Kármán constant κ. Integrating the equation one obtains logarithmic velocity profiles

$$\Phi_1(y^+) = \frac{\langle u \rangle}{u_\tau} = \frac{1}{\kappa} \ln(y^+) + A \ ,$$

$$\Psi_1(\eta) = \frac{\langle u \rangle - U_\infty}{u_\tau} = \frac{1}{\kappa} \ln(\eta) + B \ ,$$

where A and B are the integration constants.

The von Kármán constant κ and integration constants are generally considered to be universal, i.e. independent of Reynolds number and flow geometries. However, there is no generally agreed values for these constants since these constants are different from one experiment/simulation to the other, see e.g. Nagib & Chauhan (2008). Except for the log-law, there are other proposed scalings for the overlap region, e.g. a power-law description. For more discussion about power law, one may refer to George & Castillo (1997) and the references therein.

Similarly, one can also derive the law of the wall, the defect law and the log-law for the scalar field. The resulting equations read

$$\theta^+ = \frac{\theta_w - \langle \theta \rangle}{\theta_\tau} = \Phi_\theta(y^+, Pr) \ ,$$

$$\frac{\langle \theta \rangle - \theta_\infty}{\theta_\tau} = \Psi_\theta(\eta, Pr) \ ,$$

$$\Phi_\theta(y^+, Pr) = \frac{\theta_w - \langle \theta \rangle}{\theta_\tau} = \frac{1}{\kappa_\theta} \ln(y^+) + A_\theta(Pr) \ ,$$

$$\Psi_\theta(\eta, Pr) = \frac{\theta_\infty - \langle \theta \rangle}{\theta_\tau} = \frac{1}{\kappa_\theta} \ln(\eta) + B_\theta(Pr) \ ,$$

where κ_θ is the von Kármán constant for the scalar field with A_θ and B_θ being functions of Pr. θ_w and θ_∞ are the scalar concentration at the wall and in the free stream, respectively. The friction scalar θ_τ is analogous to the friction velocity u_τ and defined by

$$\theta_\tau = \frac{q_w}{\rho c_p u_\tau} \ ,$$

where ρ is the density of the fluid, c_p is the scalar capacity of the fluid, u_τ is the friction velocity and q_w is the rate of the scalar transfer from the wall to the flow which is defined by Fick's first law $q_w = -k \frac{\partial \langle \theta \rangle}{\partial y}\Big|_{y=0}$, where k is the scalar conductivity.

CHAPTER 3

Turbulent boundary layer

3.1. Mean statistics

Figure 3.1 depicts the mean velocity profiles at four different Reynolds numbers ranging from $Re_\theta = 671$ to $Re_\theta = 2512$ obtained from DNS. As the Reynolds number increases, the logarithmic region becomes much more extended and clearer. The data at $Re_\theta = 1410$ obtained by Spalart (1988) differs from the one by Schlatter et al. (2009), probably due to a suspected lower actual Re_θ in the former simulation. The excellent agreement between the simulation data and the experimental one indicates that the mean statistics are insensitive to the numerical and experimental setup. The root-mean-square (RMS) values of the scalar fluctuation are shown in Figure 3.2. With increasing Pr, the RMS values increase and the peak positions move towards the wall. Due to isoflux wall boundary condition, the RMS values remain constant near the wall while the ones with isoscalar wall boundary condition decrease to zero.

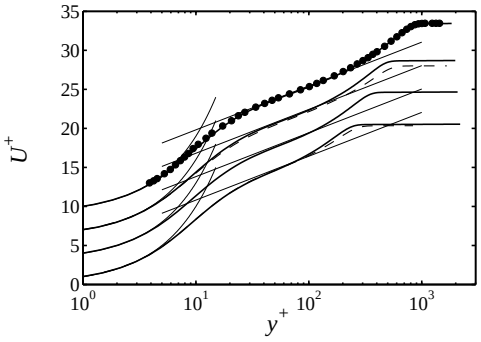


FIGURE 3.1. Mean streamwise velocity profiles U at $Re_\theta = 671, 1000, 1412, 2512$. ——DNS by Schlatter et al. (2009), • experimental data by Schlatter et al. (2009), - - -DNS by Spalart (1988). The thin lines indicate the linear and logarithmic laws, using $\frac{1}{\kappa}\ln y^+ + 5.2$ with $\kappa = 0.41$. A detailed discussion of the figure is found in Paper 2.

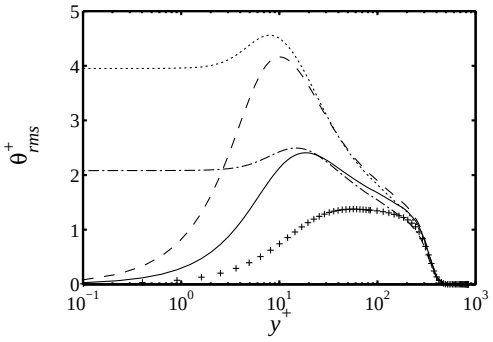


FIGURE 3.2. Profiles of the RMS values of the scalar fluctuation θ_{rms} at $Re_\theta = 830$. + θ_1 with $Pr = 0.2$, ——θ_2 with $Pr = 0.71$, –·–θ_3 with $Pr = 0.71$, – – –θ_4 with $Pr = 2.0$, ····· θ_5 with $Pr = 2.0$. Isoscalar wall: θ_1, θ_2 and θ_4. Isoflux wall: θ_3 and θ_5. A detailed discussion of the figure is found in Paper 4.

3.2. Coherent structures

Even though wall-bounded turbulence is characterised by chaotic and random motions, there are actually recurrent and well-organised features, i.e. coherent structures and motions present in the flow. Taking the definition used in Robinson (1991), a coherent structure or motion is defined as "a three-dimensional region of the flow over which at least one fundamental flow variable (velocity component, density, temperature, etc.) exhibits significant correlation with itself or with another variable over a range of space and/or time that is significantly larger than the smallest local scales of the flow."

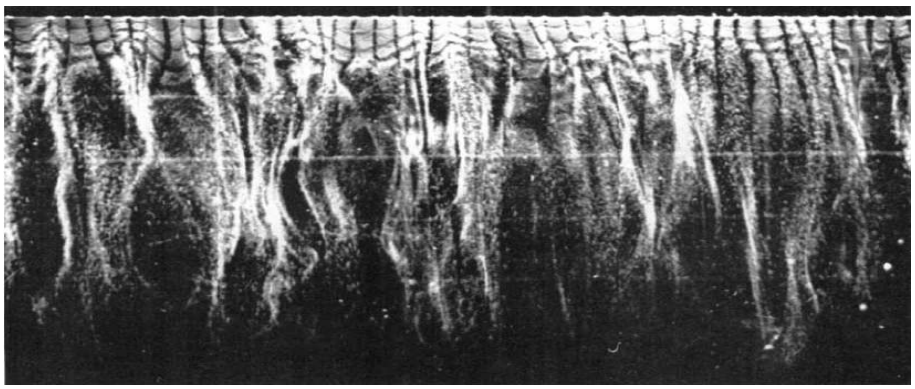


FIGURE 3.3. Photograph of the structure of a flat-plate turbulent boundary layer at $y^+ \approx 2.7$. The picture is taken from Kline *et al.* (1967).

The most widely studied structure in wall-bounded flow is the so-called near-wall streaks which are alternating high- (positive fluctuation) and low-speed (negative fluctuation) regions elongated in the streamwise direction. Figure 3.3 shows such streaky structures near the wall in a water channel taken from Kline *et al.* (1967) using tiny hydrogen bubbles released periodically from a thin platinum wire on the top. These streaky structures have a spanwise spacing of about $\Delta z^+ \approx 100$, and a streamwise spacing $\Delta x^+ > 1000$ (Smith & Metzler 1983). However, recent simulations (Li *et al.* 2009; Schlatter et al. 2009) and experiments (Lin *et al.* 2008) reported a value around 120 for the spanwise spacing of the streaks. These structures are relatively quiescent most of the time. However, the low-speed fluid near the wall occasionally erupts violently into the outer region of the boundary layer (ejection motion). Kline *et al.* (1967) and Kim *et al.* (1971) were among the first to name this process as "bursting" which was later used by Wallace *et al.* (1972), Willmarth & Lu (1972) and Lu & Willmarth (1973) among others. During a bursting process, as described by Kim *et al.* (1971), the low-speed streaks were observed first to lift up slowly away from the wall, then start a growing oscillation and finally break up into more chaotic motion. Intermittently, the high-speed fluid also rushes in toward the wall at a shallow angle (sweep motion). Together with the ejection motion, these two events contribute to most of the production of the turbulence and are considered to be a self-sustained and quasi-cyclic sequence (Robinson 1991). Jiménez & Pinelli (1999) showed that both the formation of streaks and the generation of vortices from the instability of streaks are important to maintain the turbulence.

Besides the near-wall streaks, there are also structures existing in the outer region of the boundary layer (Kim & Adrian 1999; del Álamo and Jiménez 2003; Abe *et al.* 2004; Hutchins & Marusic 2007; Schlatter et al. 2009). These

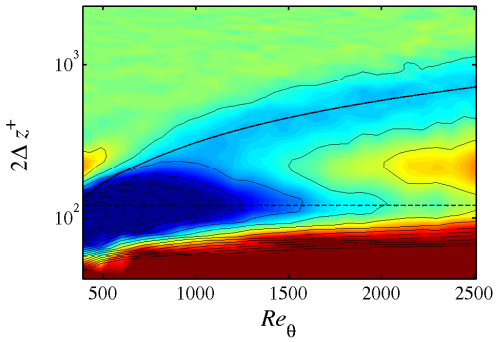


FIGURE 3.4. Spanwise two-point correlation $R_{\tau\tau}$ of the wall-shear stress τ_w. ——— corresponds to $0.85\delta_{99}$, - - - corresponds to $2\Delta z^+ = 120$. A detailed discussion of the figure is found in Paper 2.

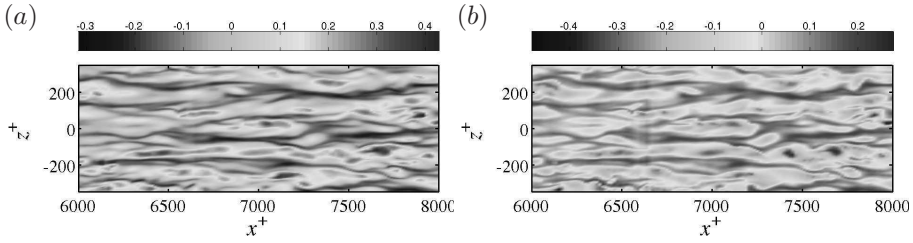


FIGURE 3.5. Instantaneous streamwise velocity and scalar fields at $y^+ \approx 7$. (a) u', (b) θ' with $Pr = 0.71$. A detailed discussion of the figure is found in Paper 4.

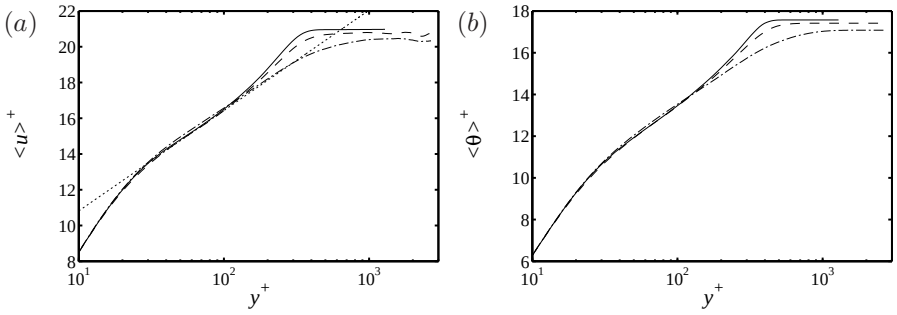


FIGURE 3.6. Mean velocity and temperature fields with FST at $Re_\theta \approx 850$. (a) $\langle u \rangle$, (b) $\langle \theta \rangle$ with $Pr = 0.71$. ——no FST, – – –$Tu = 4.7\%$ at the inlet, —·—$Tu = 40\%$ at the inlet, ····· log-law. A detailed discussion of the figure is found in Paper 5.

structures relate to the "inactive" motion by Townsend (1976) and usually scale with outer units, e.g. the channel height h or the boundary-layer thickness δ_{99}. Even though the near-wall cycle seems to be self-sustained, it is believed that the outer structures have at least a modulating influence on the near-wall events (Robinson 1991). This has been observed both in simulation and experiments, see e.g. Hutchins & Marusic (2007*b*); Schlatter et al. (2009). One such example is shown in Figure 3.4 where the spanwise two-point correlation of the wall-shear stress τ_w is shown as a function of Reynolds number. Except for the near-wall peak at $2\Delta z^+ \approx 120$ (dash line) showing the existence of the streaks, a second peak (solid line) is clearly visible for $Re_\theta > 1500$ which indicates the footprints of the large-scale structure onto the fluctuating wall-shear stress.

Kim and Moin (1989) reported that the correlation coefficient between the streamwise velocity and the scalar ($Pr = 0.71$) is as high as 0.95 in the near-wall region. Therefore, it is expected that the mean spanwise spacing of the scalar/thermal streaks is about 100 in wall units as well. Furthermore, it is also

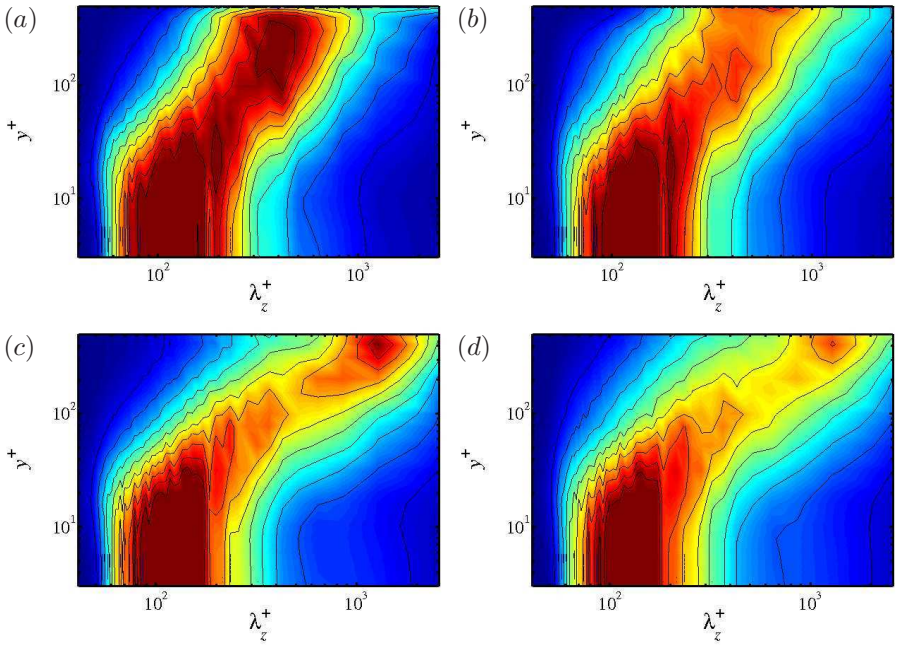


FIGURE 3.7. Premultiplied spanwise energy spectra of $\Phi_{uu}(\lambda_z)/u^2_{rms}$ and $\Phi_{\theta\theta}(\lambda_z)/\theta^2_{rms}$ with $Pr = 0.71$ at $Re_\theta \approx 900$. (a) $\Phi_{uu}(\lambda_z)/u^2_{rms}$ with no FST at inlet, (b) $\Phi_{\theta\theta}(\lambda_z)/\theta^2_{rms}$ with no FST at inlet, (c) $\Phi_{uu}(\lambda_z)/u^2_{rms}$ with $Tu = 40\%$ at inlet, (d) $\Phi_{\theta\theta}(\lambda_z)/\theta^2_{rms}$ with $Tu = 40\%$ at inlet.

confirmed that the low-speed fluids are associated with low scalar concentration regions and high-speed fluids with high scalar concentration regions (Iritani *et al.* 1985; Kong et al. 2000; Li *et al.* 2009). A visualisation of near-wall velocity and scalar streaks is shown in Figure 3.5. However, when there are disturbances present in the free stream, differences start to appear. Previous experimental work by Simonich & Bradshaw (1978); Blair (1983*a*,*b*) found that the heat transfer on the wall grows with increasing turbulence intensity in the free stream. This is very crucial for the industry application since it has direct influences on the cooling design and the hardware durability. Moreover, the wake region of the mean velocity and temperature profiles is depressed due to a high free-stream turbulence (FST) as also shown in Figure 3.6.

For the large-scale structures of the scalar without FST, Abe and Kawamura (2002) reported the existence of such structures in channel flow for $Re_\tau = 640$ with $Pr = 0.025$ and 0.71, but no detailed discussion was provided. Li *et al.* (2009) also observed the large-scale structure in the outer region of the boundary layer at $Re_\theta = 830$. But at such a relatively low Reynolds number, the outer peak in the energy spectra is not obvious due to insufficient scale

separation. With FST present, the spanwise scale of the outer structures grows from $\mathcal{O}(\delta_{99})$ to $3 \sim 4\mathcal{O}(\delta_{99})$, see Figure 3.7. It is speculated that it is these large structures which cause the increase of the heat transfer from the wall.

CHAPTER 4

Numerical method

The results presented in this thesis are obtained using the simulation code **SIMSON** (Chevalier *et al.* 2007) developed at KTH Mechanics over many years. The governing equations are discretised based on a standard spectral method (Canuto *et al.* 1988). The discretisation in the horizontal directions uses Fourier expansions assuming periodic solutions. In the wall-normal direction, Chebyshev polynomials with the Chebyshev tau method (CTM) (Canuto *et al.* 1988) are used to discretise the solution and the boundary condition. The time advancement used is a third-order Runge-Kutta method for the non-linear terms and a second-order Crank-Nicolson method for the linear terms. Adaptive time stepping is adopted to exploit the maximum stable time step. The non-linear terms are calculated in physical space rather than spectral space (pseudo-spectral method). Aliasing errors from the evaluation of the non-linear terms are removed by the 3/2-rule (Canuto *et al.* 1988) when the horizontal FFTs are performed.

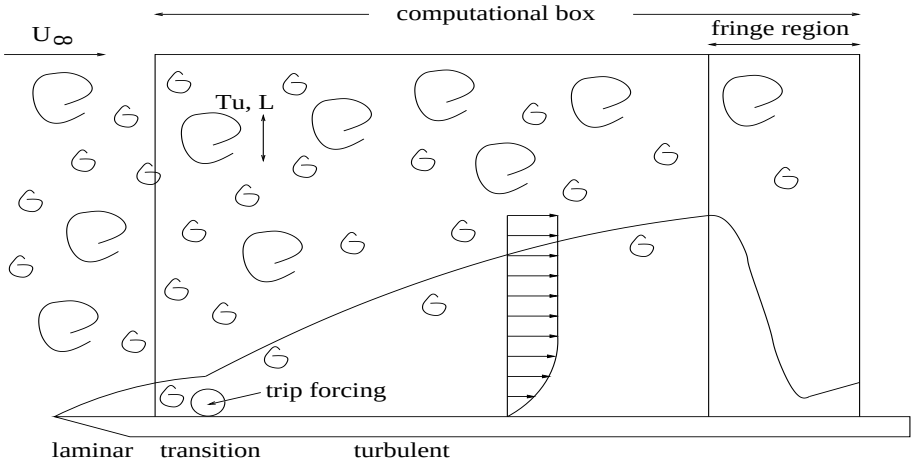


FIGURE 4.1. Sketch of the computational domain with free-stream turbulence.

A sketch of the computational box is shown in Figure 4.1. The periodicity in the wall parallel directions is fulfilled by adding a so-called "fringe region" (Bertolotti et al. 1992; Nordström et al. 1999) at the downstream end of the

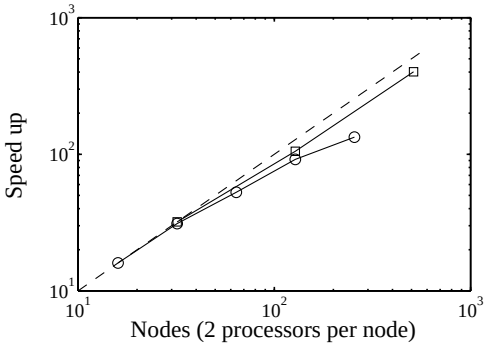


FIGURE 4.2. Speedup curve for the benchmark case. ◯ 1D parallelisation, □ 2D parallelisation, – – –linear speedup.

domain. In this region, the outflow is forced by a prescribed volume force to the laminar Blasius inflow. In addition, to trigger rapid (natural) laminar-turbulent transition, a random volume forcing directed normal to the wall is located at a short distance downstream of the inlet. The turbulent field generated by the trip force leads to very good quality at the expense of a slightly enlarged transitional inflow region. In particular, there are no streamwise correlations in the fluctuations as opposed to e.g. the rescaling and recycling method proposed by Lund *et al.* (1998).

SIMSON is also capable of simulating free-stream turbulence (FST). As described in Grosch & Salwen (1978), the continuous eigenmodes of the Orr-Sommerfeld/Squire operator provide a basis for the FST. Therefore, the generation of the FST is by a superposition of eigenmodes of the Orr-Sommerfeld/Squire operator with prescribed energy spectrum, i.e. the von Kármán spectrum. For more details about the generation of FST, one could refer to Brandt *et al.* (2004); Schlatter (2001); Jacobs & Durbin (2000). The FST is homogeneous and isotropic, and satisfies the continuity constraint as well.

The numerical code is written in FORTRAN 77/90 and consists of two major parts: one linear part where the equations are solved in spectral space, and one non-linear part where the non-linear terms are computed in physical space. The main computational effort in these two parts is in the FFT which consumes more than 55% of the total execution time. Thanks to the parallelisation, the expensive computation of the FFTs is not the main problem; the communication between processors being the bottle neck. The present code has been highly parallelised either using OpenMP (Multi Processing) on shared memory machines or MPI (Message Passing Interface) on distributed memory machines. Recently, a new parallelisation has been implemented in which the data is distributed in a "pencil-based" 2D manner instead of the previous "slice-based" 1D way. Figure 4.2 shows a performance of the benchmark case ($512 \times 513 \times 512$) and very good speedup of the code is obtained.

CHAPTER 5

Summary of the papers

Paper 1

Spectral simulations of wall-bounded flows on massively-parallel computers.
In order to take advantages of the large number of processors which the massively parallel computer systems usually have, a new 2D parallelisation of the code **SIMSON** is introduced in this report. The performance of the major subroutines is analysed in detail to identify the bottle-necks of the code. The overall performance of the new 2D version is promising. In addition, a performance model is proposed and it predicts the performance well for the benchmark cases.

Paper 2

Turbulent boundary layers up to $Re_\theta = 2500$ *studied through simulation and experiment.*
Twenty years passed since the first numerical simulation of turbulent boundary layer was done. However, the Reynolds number achieved then is still among the highest despite the rapid development of super computers and algorithms. This article is about a joint experimental/DNS work of a flat-plate turbulent boundary layer aiming at providing the research community with reliable data for a truly spatially developing flow. It is the fist time that the Reynolds number is high enough to compare DNS to carefully obtained experimental data. On the other hand, it is also a big challenge for the experimentalists to conduct an experiment at such a low Reynolds number with good quality. The excellent match of the data obtained from both simulation and experiments indicates that quantities of interest are not sensitive to the experimental and numerical limitations.

Paper 3

Simulations of high-Reynolds number turbulent boundary layers.
This paper is a continuation of the previous work of a flat-plate turbulent boundary layer using LES technique reaching the high $Re_\theta = 4300$. A promising LES model is used with comparably good resolution and the results obtained at lower Reynolds number $Re_\theta = 2500$ compare very well with the DNS data. With the high Reynolds number reached in this study, the scale separation is much more clearer and the outer structures are studied through

one-dimensional spanwise and temporal premultiplied spectra. The coherent structures observed in the simulation compare very well with previous experimental studies at similar Reynolds number.

Paper 4

DNS of a spatially developing turbulent boundary layer with passive scalar transport.
A turbulent boundary layer is simulated together with five different passive scalars upto $Re_\theta = 830$. The focus is on the outer region behaviour of the scalar statistics which is distinct from channel flows. The influences of the different boundary conditions and Prandtl numbers are also discussed. With such a variety of scalars, the Reynolds number reached in this study is the highest in the current literature.

Paper 5

Simulations of heat transfer of a boundary layer subject to free-stream turbulence.
This proceeding article is a continuation of the work presented in Paper 4 but with additional influence of the free-stream turbulence. It has been long time that researchers have observed that in the presence of free-stream turbulence, the heat transfer on solid walls increases dramatically. Since there are difficulties in experiments to measure very close to the wall, numerical simulations become a good alternative to shed light on the physical mechanisms. The present study summarises the previous experimental and simulation results, and furthermore attributes the increase of the heat transfer on the wall to the large-scale structures residing in the outer region of the boundary layer.

CHAPTER 6

Conclusions and outlook

In the present thesis, spatially evolving turbulent boundary layers together with passive scalars are investigated numerically. The research code is parallelised so that it can run on more than 1000 processors and this together with the quickly developing parallel computer systems opens the possibility to study wall turbulence at numerically very high Reynolds numbers.

The close agreement with the experimental data at $Re_\theta = 2500$ increases the credibility of obtained statistics at the low to medium Reynolds numbers. It further validates that the results are insensitive to the details of the experimental and numerical setup, e.g. the surface roughness, tripping device, streamwise pressure gradients, boundary conditions. By analysing the spanwise and temporal spectra, the large-scale structure is found to be about $0.85\delta_{99}$ in width and persists for about $10\delta_{99}/U_\infty$ time units.

For the results pertaining to heat transfer, the mean scalar profiles are independent of the different boundary conditions whereas the effects on the scalar variances and high-order statistics are obvious in the near-wall region. The scalar with isoscalar boundary condition is highly correlated with the streamwise velocity component in the near-wall region. However, near the boundary-layer edge, only a mild correlation between these two quantities was observed. With the presence of free-stream turbulence, a more prominent depression of the wake region can be observed with increasing turbulence intensity for the mean velocity and temperature profiles. As much as 15% increases of both the skin-friction coefficient and the Stanton number are found for a turbulence intensity of 20% at the inlet.

Therefore, a natural extension of the present work is to simulate such canonical flows at even higher Reynolds numbers since most engineering applications happen at very high Reynolds numbers. With the present increasing speed of the computer power, by the middle of 21 century, to simulate the wing of an airplane will not be a dream anymore. Another possibility is to study the vortical structures residing in the outer layer of the boundary layer in details. One and two-dimensional spectra will be analysed and visualisations of those structures will be done, see also Figure 6.1.

For the SGS modelling, the various models used in the present work have been shown to be very efficient and accurate. Recently, a new variant of mixed model, e.g. HPF-WALE model, also shows promising results in channel flows.

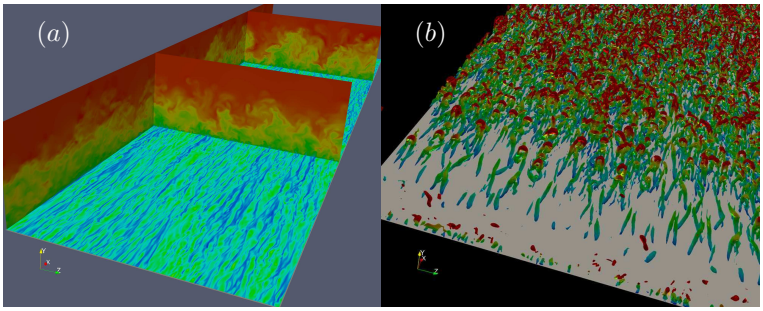


FIGURE 6.1. (*a*) A visualisation of an instantaneous flow field at $Re_\theta \approx 4000$. (*b*) Iso-surfaces of the second invariant of the velocity gradient tensor Q. See also Wu & Moin (2009).

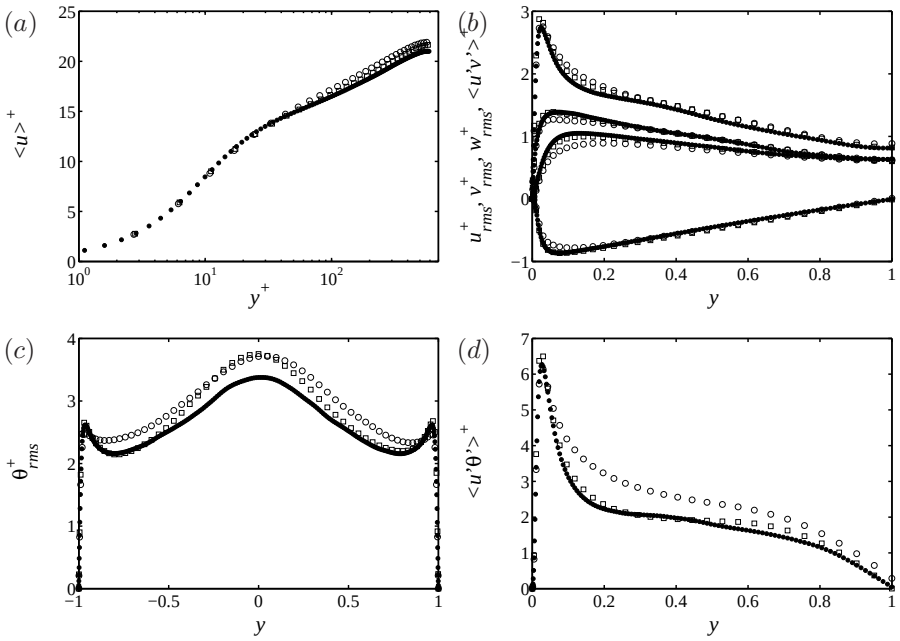


FIGURE 6.2. LES results of channel flow with passive scalar ($Pr = 0.71$) at $Re_\tau = 590$. • DNS (unfiltered), ∘ DSM model, ▫ HPF-WALE model.

Extensions of these models including passive scalar are interesting. Preliminary results of channel flow simulations with heat transfer ($Pr = 0.71$) at $Re_\tau = 590$ are shown in Figure 6.2 with comparison to the results obtained using DSM model and DNS data. In general, the results obtained with the new HPF-WALE model outperform those from DSM model. Both models overpredict the

mean velocity and scalar variance profiles in the centre of the channel. However, the resolved stresses and scalar fluxes predicted by the HPF-WALE model are in good agreement with the unfiltered DNS data. Further validation of these models will be performed in zero-pressure gradient turbulent boundary layers since whether the LES models are able to reproduce the high Reynolds number effects in turbulent boundary layer flows or not is still an open question. Other alternatives will be to investigate LES models which account for anisotropy, e.g. the explicit algebraic Reynolds stress model (EARSM) developed by Marstorp (2008). It is also interesting to combine the stochastic modelling approach with the available models. Furthermore, to test the SGS models in non-equilibrium situations and flows with separation induced by adverse pressure gradient would also be interesting.

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Gan xie jia ren he qin qi peng you dui wo de guan huai!

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Part II

Papers